
FINE-TUNING GENEALOGY IS WRITTEN IN WEIGHTS

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ABSTRACT

1 INTRODUCTION

Write introduction.

2 RELATED WORK

Write related work.

3 FRAMEWORK

3.1 PHYLOGENETIC TREE RECOVERY

Write phylogenetic framing.

3.2 DISTANCE METRIC: COSINE AND THE SNR ARGUMENT

Write metric selection rationale.

3.3 LAYER-LEVEL TREES AS INDEPENDENT ESTIMATES

Write layer ensemble framing.

4 EXPERIMENTS

4.1 CONTROLLED BRANCHING TREES

Write experiment setup.

4.2 GEOMETRIC MECHANISM: TASK SUBSPACES

Write geometric mechanism.

4.3 LoRA VARIANTS AND FULL FINE-TUNING

Write LoRA experiment description.

4.4 BEHAVIORAL BRIDGE

Write behavioral bridge experiment description.

4.5 BLACKBOX: FRONTIER MODEL FAMILIES

Write blackbox experiment description.

Table 1: Tree recovery accuracy by tree size (branching_v3, cosine distance). Observed RF is mean $\pm$ SE across runs; null mean from 1,000 random binary trees. % sig: fraction of runs with $p < 0.05$ vs. random null. Fisher p : combined significance across runs.

Leaves	n	Null mean (SE)	Observed RF (SE)	% sig	Fisher p
5	11	3.21 (0.02)	0.55 (0.21)	0%	0.003
6	12	5.24 (0.02)	1.92 (0.36)	50%	0.0003
7	15	7.33 (0.01)	2.60 (0.29)	60%	<0.0001
8	4	9.42 (0.03)	3.75 (0.48)	100%	<0.0001



Figure 1: RF distance by tree size for whitebox weight distances (cosine, blue triangles), behavioral distances (RTE label agreement, red squares), and random null (gray band, P10–P90). Whitebox sits well below the null at all tree sizes ≥ 5 leaves; behavioral distances are indistinguishable from random. Behavioral failure is the point: it confirms that weight access is necessary for per-run lineage recovery.

5 RESULTS

5.1 WEIGHTTREE: LINEAGE IS RECOVERABLE FROM WEIGHTS

Write WeightTree results prose. Headline: 60% sig at 7 leaves, 100% at 8. Per-run significance is the operational claim. Ref Figure 1 and Table 1.

5.2 LAYERTRACE: SIGNAL CONCENTRATES IN 4% OF THE MODEL

Write LayerTrace results prose. enc_sak (4% of params) non-inferior to full weight at all tree sizes ≥ 5 leaves; outperforms at 6 leaves (RF 1.38 vs 1.92). Ref Figure 2 and Table 2.

5.3 SUBSPACE ANGLES AND GEOMETRIC MECHANISM

Write subspace angles results.

5.4 LORA DOSE-RESPONSE

Write LoRA dose-response prose. Key sentence: lineage survives LoRA at all. Large gap between full_ft_approx and full FT = constrained vs unconstrained updates. Ref Table 2.

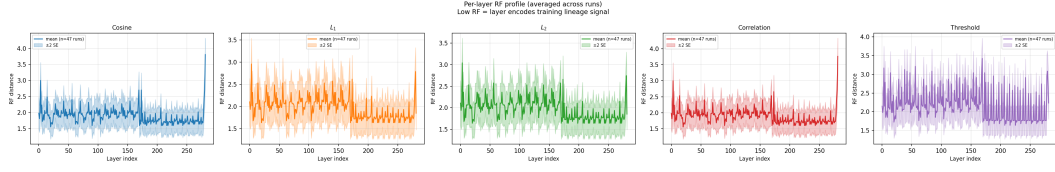


Figure 2: Per-layer RF distance (mean $\pm$ 2 SE, $n = 47$ runs) for all five distance metrics. Encoder SA.K layers (highlighted) cluster at the bottom across all metrics — consistently lowest RF, strongest lineage signal. The cyclical pattern (attention $\rightarrow$ FFN $\rightarrow$ layer norm repeating across blocks) shows that lineage concentrates in attention keys and dissipates through the FFN.

Table 2: Lineage signal by training regime (branch structure analysis, cosine distance, TRAINED_ONLY=1 for LoRA groups). Fisher- z r : Fisher z -averaged within-run correlation between cosine distance and same-branch indicator. % sig: runs with $p < 0.05$.

Variant	Trained layers	Fisher- z r	SD	N runs	% sig (7-leaf)
<i>LoRA adapter scope (Flan-T5-base)</i>					
k_only	36	-0.485	0.330	37	—
qkv	36	-0.594	0.250	36	58%
full_attn	144	-0.671	0.245	32	73%
full_ft_approx	168	-0.687	0.242	31	70%
<i>Full fine-tuning</i>					
branching_v3	all	-0.805	0.230	35	60%
<i>LayerTrace variants (full FT, 6-leaf runs)</i>					
enc_sak (4%)	12 tensors	RF 1.38 ± 0.18 vs full 1.92 ± 0.36			
full weight	282 tensors	RF 1.92 ± 0.36 (reference)			

5.5 BEHAVIORAL BRIDGE

Write behavioral bridge result (one paragraph).

5.6 BLACKBOX: FRONTIER MODEL FAMILIES

Write blackbox results prose. Key points: Claude siblings tightest; Gemini-Llama proximity = formatting artifact; translation cleanest; multi-task aggregation necessary. Ref Figure 3.

6 DISCUSSION

6.1 THE PROVENANCE CHAIN

Write provenance chain summary.

6.2 WHY ATTENTION KEYS?

Write mechanistic interpretation.

6.3 WHITEBOX VS. BLACKBOX: COMPLEMENTARY ROLES

Write whitebox vs blackbox discussion.

6.4 LIMITATIONS AND FUTURE DIRECTIONS

Write limitations and future work.

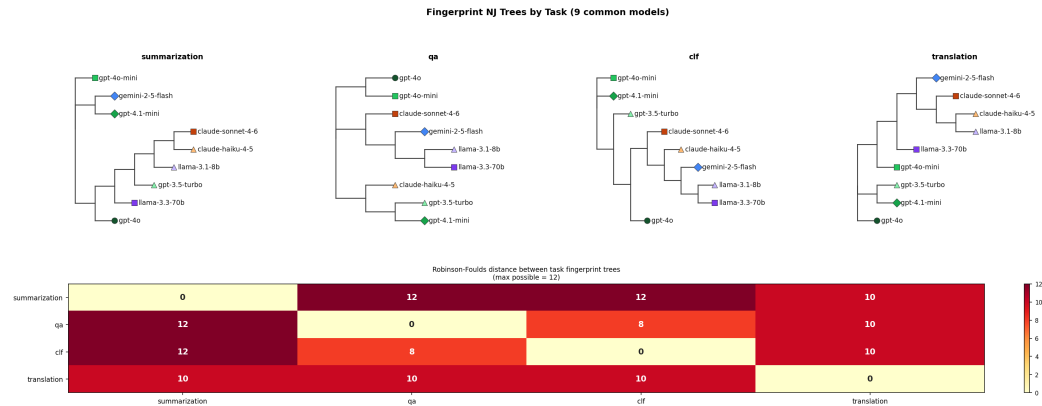


Figure 3: Neighbor-joining trees from behavioral fingerprint distances, one panel per task. GPT models (green) cluster consistently across all tasks. Claude clusters in summarization and QA but splits in translation (Haiku migrates due to its unique header-heavy formatting style in that task). Gemini–Llama proximity in QA and classification is a formatting artifact: both produce verbose, structured responses on those tasks, while GPT gives terse label-only answers. The artifact attenuates in summarization, where response length is task-constrained.

ACKNOWLEDGMENTS

Add acknowledgments for camera-ready.

A EXPERIMENTAL DETAILS

Fill experimental details.

B LAYER VARIANT DEFINITIONS

Fill layer variant table.

C NULL DISTRIBUTION CONSTRUCTION

Fill null distribution description.

D BLACKBOX EXPERIMENT DETAILS

Fill blackbox details.

E METRIC SELECTION: SNR ANALYSIS

Write SNR analysis text.

F BEHAVIORAL REGRESSION: FULL RESULTS

Fill regression appendix.

G WITHIN-TASK AND ROBUSTNESS RESULTS

Fill robustness appendix.

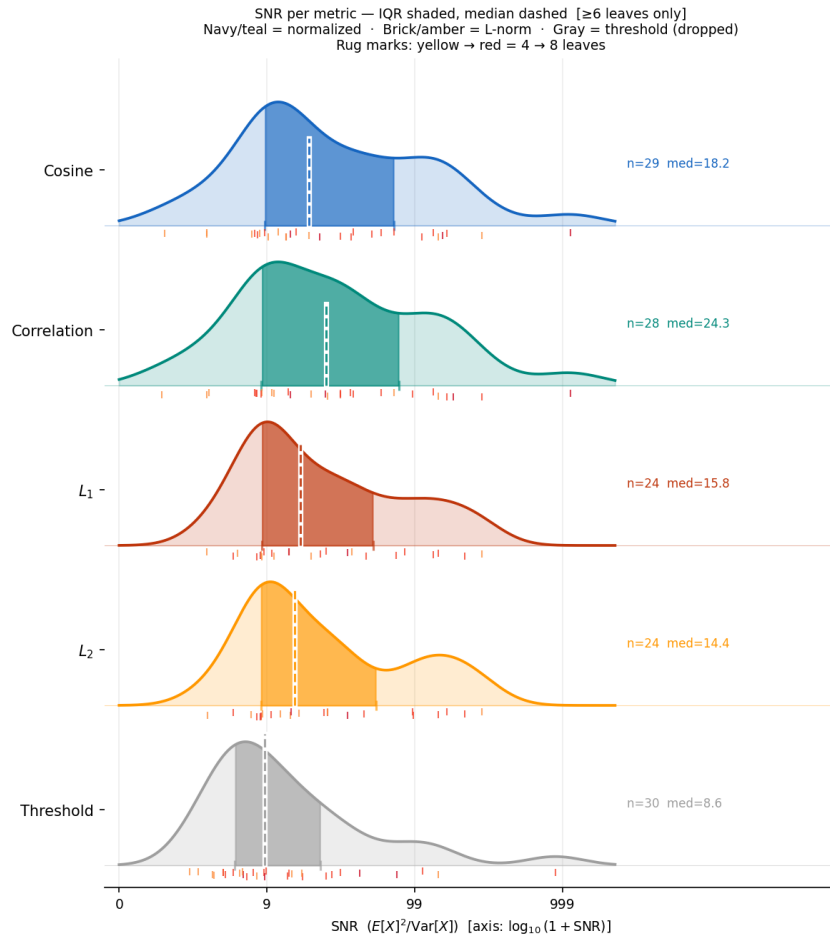


Figure 4: SNR distributions by metric family (≥ 6 -leaf runs). Normalized distances (cosine, correlation) dominate. See §3.2.

H LLAMA-3.1-8B EXPERIMENT DETAILS

Fill Llama appendix.